

The speed of sound in a two-dimensional hard-disk gas: a divider-mode measurement against the equation of state

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Abstract—We measure the adiabatic speed of sound of a hard-disk fluid from the eigenfrequency of a massive divider separating two compartments, the method of Román *et al.*, over 35 packing fractions from $\eta = 0.0065$ to 0.76 and nine divider masses per density. The measurement is compared with the Kolafa-Rottner equation of state through the adiabatic relation $c_s^2 = (k_B T/m)[Z + \eta Z' + Z^2]$ with no fitted parameter. For $\eta \leq 0.4$ the agreement is $+1.04\%$ on average at $N = 100$; a size ladder to $N = 2500$ shows the offset is finite size. We bound the estimator's own systematic below 0.3% , and we show that the apparent density-dependent excess reported in 2002 is an artefact of the $\gamma = 2$ mapping used then: re-mapped adiabatically, that 24-year-old data set shows the same flat $+1.1\%$ offset as ours, in the same box.

Index Terms—hard-disk fluid, speed of sound, event-driven molecular dynamics, equation of state, finite-size scaling

I. INTRODUCTION

THE hard-disk fluid is the simplest system with a non-trivial equation of state, and its thermodynamics is known to high precision from simulation [1]. Its *dynamical* response is measured less often. The speed of sound is the natural bridge: for a two-dimensional monatomic fluid it is fixed entirely by the equation of state,

$$c_s^2 = \frac{k_B T}{m} [Z + \eta Z' + Z^2], \quad (1)$$

so a measurement of c_s tests $Z(\eta)$ with the equation of state as an *input* rather than a fit. Agreement is then a check; disagreement is a measurement of something the equation of state does not contain.

Measuring c_s by launching a pulse needs a large box and care with the pulse amplitude. Román *et al.* [2] avoid both by placing a massive divider between two compartments and watching it ring: the divider is a mass on two gas springs and its eigenfrequency is set by c_s . The measurement becomes a frequency, which is cheap to make precise, and the system can be small. We adopt that apparatus, and much of this paper is about what it takes to trust the frequency it returns.

Two things motivate revisiting it. First, the estimator. The divider carries a second, much slower motion that is not the sound mode, whose spectral weight can exceed the resonance;

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Fig. 1. The apparatus, drawn by the simulation itself. Two compartments of 50 disks either side of the divider. Walls are labelled by mass; a held wall is drawn solid and a free one hollow. The wall at left bounds an inert compartment belonging to a companion experiment and is held immovable throughout this work.

an unguarded peak-finder is wrong by tens of per cent (§III). Second, the comparison. Román *et al.* convert Z to c_s with a constant $\gamma = 2$ and report an excess over theory that grows with density; we show in §VII that the growth is the mapping, and that their data and ours then agree.

II. MODEL AND APPARATUS

Hard disks of diameter $\sigma = 2r$ and mass m occupy a box of height H split at the centre by a divider of mass M and thickness t , with N_s disks on each side. Units are reduced, $k_B T = m = \sigma = 1$; we use $H = 10$ and $N_s = 50$ unless stated.

The dynamics are event-driven and exact between collisions; all results use the scientific backend. The divider is held for an equilibration period, during which each compartment is separately equalised to $KE = N_s k_B T$, and is then released; its position $x(t)$ is recorded at fixed sampling interval. Every trajectory is checked against a health contract — a scheduler forced advance, an overlap repair, a wall-position clamp or an overdue wall collision each discard that trajectory, never the cell. Of 7875 trajectories in the main campaign, 2 were discarded.

a) *Effective length*: The disk centres in one compartment span r to $L_0 - t/2 - r$, so the length entering the mode is

$$L_{\text{eff}} = L_0 - 2r - t/2. \quad (2)$$

Carrying t lowers every c_s by $(t/2)/(L_0 - 2r)$: 0.004% at $\eta = 0.0065$ and 0.50% at $\eta = 0.65$. The density is defined on the physical area, $\eta = N_s \pi r^2 / (L_0 H)$, with no exclusion — L_{eff} is an acoustic length and enters nowhere else.

III. THE ESTIMATOR

For each trajectory we form the periodogram of the divider position,

$$P(f) = \left| \sum_j (x(t_j) - \bar{x}) e^{-2\pi i f t_j} \right|^2, \quad (3)$$

and take the largest bin at or above a one-sided floor,

$$\nu = \arg \max_{f \geq \nu_{\text{pred}}/X} P(f), \quad X = 2.5, \quad (4)$$

then average over 25 seeds. With record length T the bin width is $1/T$, so at 200 predicted periods the rounding floor on one frequency is $100/(N_{\text{cyc}}\sqrt{12}) = 0.144\%$.

a) Why a floor is needed: Besides the resonance the divider performs a slow wander: a five-period running mean of x carries 9–57% of the variance of x . It is not a start-up transient — its variance is the same in the second half of a record as the first, and a hundredfold longer hold does not change it — and it is a relaxation rather than an oscillation, with a correlation time of 53–102 acoustic periods. It is the adiabatic-piston mechanism: thermal fluctuations of the two compartment temperatures shift the pressure balance, the divider follows, and it returns only as heat crosses the divider. A dedicated run logging the compartment kinetic energies confirms it directly, with correlation $r = +0.978$ between the slow mode and the temperature difference and the amplitude reproduced to 6–7% by $x = (L_0/2)(\Delta T/T) Z/(Z + \eta Z')$. Its power sits at low frequency, and at $\eta \geq 0.52$ it reaches 85–280% of the resonance peak for the lightest divider.

The consequence is quantitative. Against a reference that windows the spectrum around the resonance, an unguarded largest bin is wrong by up to 33% on 25-period records and up to 93% at 200 periods — the error grows with record length, because a longer record resolves the wander into more bins, any of which can outgrow the resonance. Short records hide the problem rather than remove it. The floor removes it, and the answer does not depend on where the floor is placed: every X from 2.08 to 6.25 gives the identical c_s at 200 periods.

b) Largest bin or a fit?: A resonance fit to each trajectory is the obvious alternative and the more precise one. Both put the nine masses of §IV on a single line to 0.27%. But a biased estimator does not scatter about the line, it leans: regressing the residual on $\ln \alpha$ gives $-0.286 \pm 0.020\%$ per e-fold for the fit (14.4σ) against $-0.030 \pm 0.029\%$ for the largest bin (1.0σ , flat). The fit walks monotonically from +0.35% at the lightest divider to -0.64% at the heaviest. We use the largest bin and report the fit as a cross-check; no equation of state entered that choice.

c) Bounding the rest: Damping shifts the position-spectrum peak below the eigenfrequency by $\Gamma^2/4f_0^2$; with the measured $\Gamma/f_0 = 0.023$ (median, $\eta < 0.5$) and 0.063 ($\eta \geq 0.5$) that is at most 0.34%, typically 0.01–0.11%. Weighting by $(2\pi f)^2$ to suppress the slow mode without a floor was tested and rejected: it moves c_s by +0.02 to +3.3%, one-signed and largest at high density, which is a spectral-shape systematic and not noise. Taken together the estimator's own systematic is below 0.3%, against the 1–3% effects the paper reports.

IV. THE MASS LADDER

Linearising about the symmetric position makes the divider a mass on two gas springs. Writing the displacement field in each compartment as a standing wave of wavenumber k , imposing continuity at the divider faces and Newton's law for the divider, $M\ddot{\xi} = -[p_{\text{right}} - p_{\text{left}}]H$, gives with $K = kL$

$$\cot K = \alpha K, \quad \alpha = \frac{M}{2N_s m}, \quad (5)$$

whose lowest root fixes the fundamental,

$$\nu = \frac{c_s K(\alpha)}{2\pi L_{\text{eff}}}. \quad (6)$$

Two limits are worth recording. For a heavy divider $K^2 \rightarrow 2N_s m/M$, so the divider becomes a mass on a spring of stiffness $N_s m c_s^2/L^2$ per compartment — the gas spring is the sound speed. For a light divider $K \rightarrow n\pi/2$ with n odd; the even standing modes are absent because they have a node where the divider sits.

Equation (6) is linear in $K(\alpha)$ through the origin, which makes the mass ladder a self-contained consistency test: a single c_s must place every mass on the same line, with no equation of state involved. We use nine masses, $M = 50$ to 2000, and take c_s as the through-origin slope of $\bar{\nu}_M$ against $K(\alpha)/2\pi L_{\text{eff}}$, its uncertainty being the scatter of the per-mass values about that line. That scatter, 0.27%, is within a factor two of the bin-rounding floor.

V. RESULTS

Over $\eta \leq 0.4$ the measurement sits +1.04% above Kolafa–Rottner on average at $N = 100$ and never worse than 1.68%. The offset is one-signed and grows slowly with density.

VI. FINITE SIZE

Along $N = 100, 400, 900, 1600$ (and 2500 where available) at fixed η , $c_s(N)$ is fitted with $c_\infty + b/\sqrt{N}$, with b/N and the two-term form bracketing the systematic. At $\eta = 0.02$, $c_\infty = 1.4614 \pm 0.0102$ against 1.4725 from the equation of state (-1.1σ); at $\eta = 0.05$, 1.5699 ± 0.0029 against 1.5666 ($+1.1\sigma$). The $N = 100$ offset closes as the box grows: it is finite size, not a defect of the method.

a) What the offset is not: The pressure measured at the wall of the same finite box exceeds the bulk equation of state by 7–20%, rising with η . Propagated through (1) that would imply a +5 to +16% excess in c_s , five times what is observed. The pressure of a finite box and its sound speed are therefore two different finite-size quantities, and the bulk relation between them does not hold at $N = 100$. We report this as a negative result: the wall pressure cannot be used to explain the offset.

VII. COMPARISON WITH ROMÁN *et al.* (2002)

Román *et al.* [2] measured the same observable in the same box ($N_s = 50$, $H = 10$) and report c_s above scaled-particle theory and Henderson by an amount that grows with density, from +1.3% at $\eta = 0.112$ to +12.8% at 0.524. That growth is the mapping. They convert Z to c_s with a constant $\gamma = 2$; using (1) on their own numbers gives Table I.

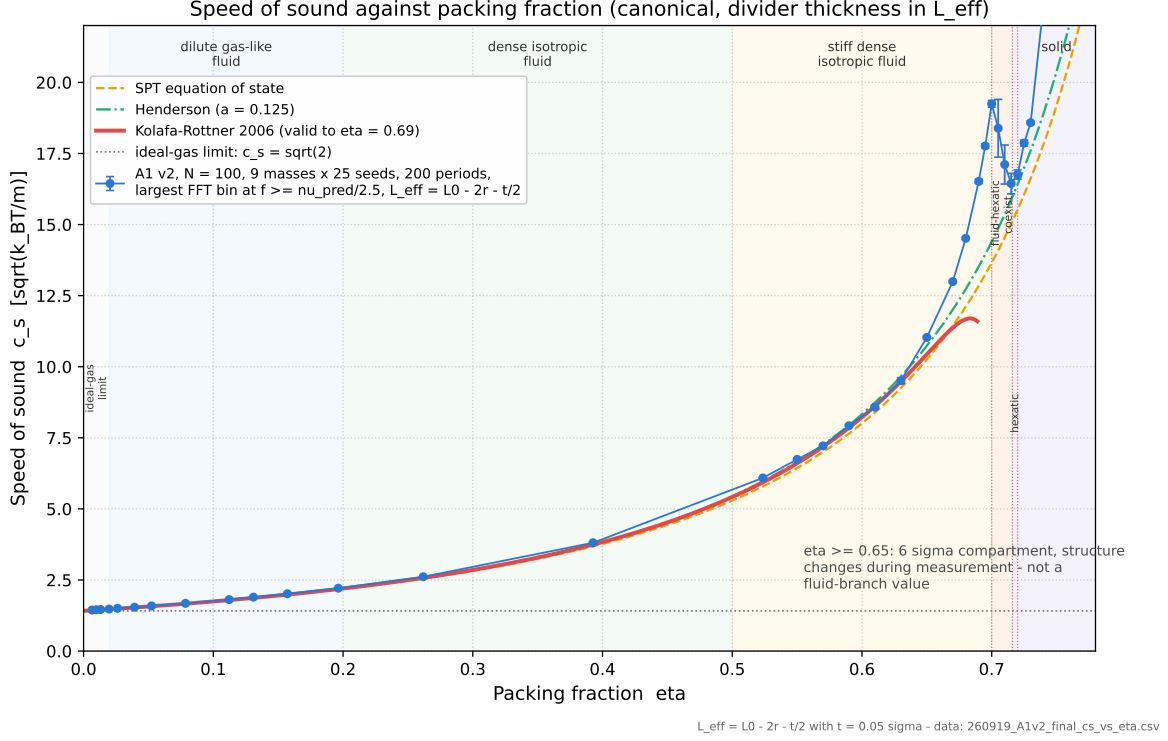


Fig. 2. $c_s(\eta)$ at $N = 100$ against three equations of state, through (1) with no fitted parameter. Shading marks the approximate regimes. Points beyond $\eta \approx 0.69$ lie in the coexistence and ordering region where no fluid reference exists; they are shown but are not counted as a deviation from Kolafa–Rottner.

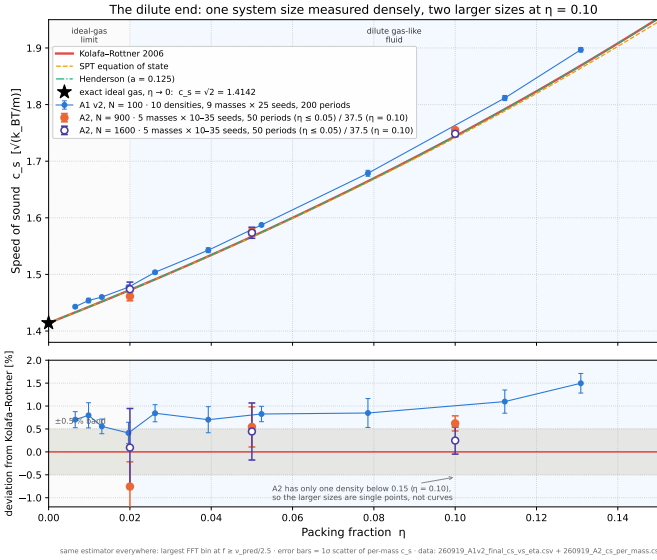


Fig. 3. The dilute end. The $N = 100$ curve sits 0.5–1.5% above the equation of state; larger systems agree with it within their errors.

The density-dependent excess becomes a flat +1.10% (spread +0.91 to +1.46%), against our +1.04% at $N = 100$ averaged over $\eta \leq 0.4$. Two independent implementations, 24 years apart, in the same box, with the same offset: that makes the +1% a property of the $N = 100$ divider apparatus rather than of either code. Their own size series (their Table II, $N = 64 \rightarrow 4096$, $c_s = 3.81 \rightarrow 3.75$, extrapolating to 3.74) runs in the same direction as ours. As a transcription check,

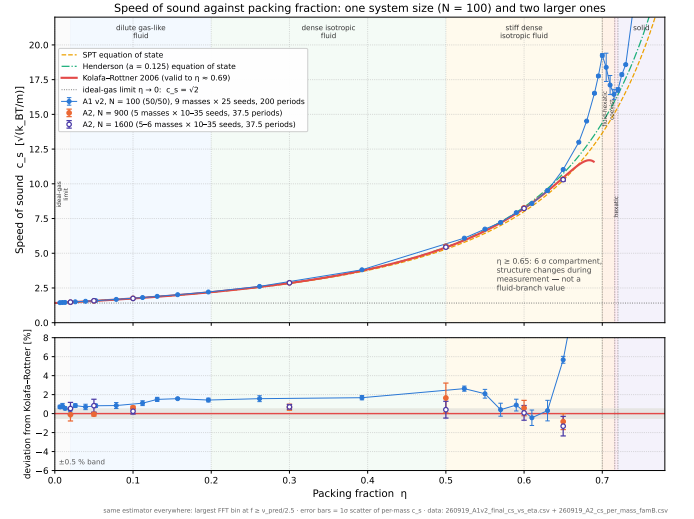


Fig. 4. $N = 100$ against $N = 900$ and 1600 at the same densities.

their scaled-particle and Henderson columns are reproduced to every printed digit by our equation-of-state code together with their $\gamma = 2$ mapping.

The two measurements differ at one point, $\eta \approx 0.52$: ours sits +2.6% and theirs $+1.0 \pm 1.5\%$, which is 1.1σ on their error bar alone, with 100 trajectories against our 25.

VIII. OUTLOOK

Two measurements remain. A size ladder at $N \approx 1000$ across the full density range would turn §VI’s two-point

TABLE I
ROMÁN *et al.* TABLE I RE-MAPPED. η IS RECOMPUTED FROM THEIR L_0 .

L_0	η	c_s (theirs)	dev., $\gamma = 2$	dev., adiabatic
7.5	0.5236	5.99 ± 0.09	+10.6 %	+1.0 %
10	0.3927	3.78 ± 0.08	+5.5 %	+0.9 %
15	0.2618	2.61 ± 0.03	+3.3 %	+1.5 %
20	0.1963	2.20 ± 0.02	+1.9 %	+0.9 %
25	0.1571	2.01 ± 0.02	+1.9 %	+1.3 %
30	0.1309	1.89 ± 0.02	+1.5 %	+1.1 %
35	0.1122	1.81 ± 0.02	+1.3 %	+1.0 %

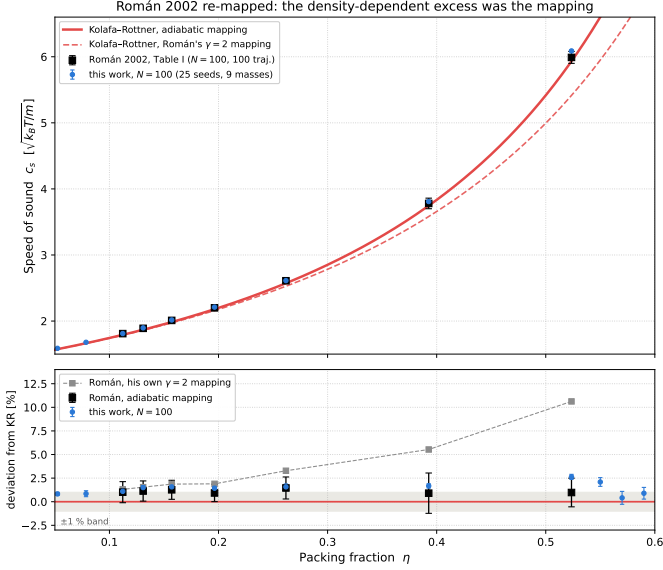


Fig. 5. Román *et al.* re-mapped. Solid red, Kolafa–Rottner through (1); dashed red, the same equation of state through their $\gamma = 2$ mapping; black squares, their Table I with their error bars; blue, this work at $N = 100$. Lower panel: deviation from Kolafa–Rottner.

extrapolation into a curve; and a size series at $\eta = 0.65$ would test whether the offset continues to close where the compartment is only six disk diameters wide. Both are cluster jobs. Above $\eta \approx 0.69$ no fluid reference exists and a global equation of state with hexatic and solid branches [3] would be needed to say anything quantitative; we show those points but make no claim about them.

REPRODUCIBILITY

Every campaign directory carries the exact command and the estimator geometry line. The geometry constant and the estimator share one definition, and the analysis refuses to run on data whose recorded wall thickness disagrees with it.

REFERENCES

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